

# Orienting Participatory AI for Frontline Care Work: A Conceptual Framework for Technologists

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Technologists increasingly turn to artificial intelligence (AI) to support frontline care workers (FCWs), via work-tracking applications, well-being monitoring systems, and other data-driven interventions aimed at improving care quality and working conditions for this vital labor and labor force. Thoughtful application of AI in this dynamic and high-stakes domain requires centering FCWs' voices in technology design and development—but doing so challenges existing approaches in participatory design (PD). For example, care work's relationality defies standardization and quantification. Moreover, it can be difficult to balance all stakeholders' needs in a hierarchical and high-stakes collaboration environment as FCWs often prioritize patients' or clients' needs above their own. This paper draws together scholarship in CSCW and participatory AI, alongside the authors' extensive experience developing AI systems in frontline care contexts, to present a conceptual framework for meaningfully engaging frontline care workers' visions of data and AI systems. We illustrate our framework through case studies of two particularly urgent applications: shift matching in home care and identifying burnout in the medical workforce. We discuss tensions in implementing this framework, including how to do equitable PD, and close with a research agenda for CSCW as the design space around AI in frontline care work expands.

CCS Concepts: • **Human-centered computing** → **HCI theory, concepts and models**.

Additional Key Words and Phrases: participatory design, AI ethics, care labor, toolkit/guidelines

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## 1 Introduction

Frontline care work is increasingly recognized as essential work. The COVID-19 pandemic demonstrated our reliance on critical services like healthcare (e.g., nursing), community and social services (e.g., counseling), and personal care services (e.g., childcare), especially as large swathes of these workforces bore the brunt of COVID's effects [130]. Additionally, many countries, including the United States and Japan, are facing an impending “*care crisis*” as the proportion of their populations aging out of the workforce and needing care are increasing, even as the proportion of young and healthy adults decreases, and family and friend care networks erode [39].

Enter frontline care workers (FCWs). We define FCWs as **paid professionals who specialize in directly supporting the well-being of another person**, drawing on discourse from CSCW literature on frontline [120] and care [66] work, as well as foundations in sociology [45, 135] and

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science and technology studies [30]. FCWs have domain-specific skills for which they receive training and for which there are standards. Their work often involves what Pugh [109] calls connective labor: the work of seeing and being with someone else. Often, frontline care work is also built upon an ethos of service, which can result in the worker putting their client's needs above their own. FCWs bear the burden of filling the gaps in our social safety net created by the care crisis. However, to date, their work has been underpaid, devalued, and frequently rendered invisible [56].

To make easier the vital human work of frontline care, technologists have sought to apply artificial intelligence (AI) and other data-driven technologies to this dynamic and high-stakes problem space. Robots are being developed to support the physical and emotional care of older adults [147], smartphone applications are being used to reduce fraud and waste in home and community-based care [84], and telehealth has emerged as a tool for connecting a strained workforce of care providers with a burgeoning population of people seeking support [71]. However, rather than improving the quality of care and enabling caregiving for even the most remote settings, many times, these systems are perpetuating the care crisis and introducing new concerns. FCWs trained to care for others instead spend countless hours learning, handling, debugging, explaining, and developing workarounds for the technologies they are told to use [138], resulting in lower-quality care for care recipients [95] and more strenuous working conditions for care providers [147].

When confronted with the introduction of a new technological paradigm into a complex setting of structured and cooperative work, many within CSCW would turn to participatory design (PD) as a set of methods for foreground workers' voices and challenging existing power structures. However, as longstanding discussions in PD have noted, social inequities can complicate or limit the possibility of marginalized populations meaningfully engaging in design [16, 55]. These problems are further amplified in an era of AI proliferation [22, 51, 148].

Moreover, applying PD for AI in frontline care also presents context-specific challenges. AI systems are increasingly reliant on massive amounts of data on workers' needs and actions—despite concerns regarding workers' abilities to meaningfully consent [24]. Frontline care work is relational and difficult to quantify, often involving invisible or intangible emotional labor [89], which clashes with the standardization required for AI. Additionally, the ethos of service inherent to frontline care can mean workers have trouble expressing their needs in PD settings, instead deferring to their clients' wishes [43, 136]. Furthermore, the frontline care setting complicates the basic PD question of which stakeholders should even be at the table. Care recipients can be just as impacted by invasive or ill-fitting workplace technologies as care workers; and there can also be settings where their needs do not align [84]. **Therefore, how do we orient participatory AI to meet the needs of frontline care work?**

The goal of this paper is to help guide technologists who are considering AI approaches to frontline care work to critically reflect on the role they intend AI to play and how best to incorporate the voices of FCWs. The goal is not to prescribe specific actions or provide a finite checklist, but rather, to support more expansive reflection—helping technologists assess whether their projects are well-suited to AI, ask contextually relevant questions, and identify participatory mechanisms that are appropriate for their specific settings. We synthesize the literature in CSCW and adjacent communities, including HCI, FAccT, participatory design on frontline care work, worker and workplace technologies, and participatory AI. We juxtapose the key questions raised by these literatures with our own extensive on-the-ground experiences in academic and industry settings designing and deploying AI systems in frontline care contexts, including with home- and community-based services as well as tertiary clinical settings across the U.S. and the Majority World. Through this triangulation, we develop a conceptual framework to orient technologists towards some of the

key considerations in participatory AI, and guide them to reflect on whether and how to do a participatory AI project in a given frontline care setting.

There are three main considerations in our framework, including: (1) the intended benefits and potential harms of deploying AI systems in the given frontline care context, (2) the macro-, meso-, and micro-level attributes of the context that can influence the participation of different stakeholders, and (3) appropriate participatory mechanisms through the project stages regarding the data, models, and system given these factors. We present these considerations through an interactive worksheet, which is similar to other design and project management tools leveraged by teams in industry such as LeanUX Canvas [53] and academic design toolkits such as the Situate AI Guidebook [73]. We also demonstrate how this worksheet could be applied to real-world examples of frontline care work contexts from our research where AI systems are being considered. Finally, we explore some of the tensions raised by applying the framework in our example case studies, such as: quantifying care, adding workload to overburdened workers, and building accountability in hierarchical and interdependent structures.

Our research contributes to the broader conversations about participation and AI while being grounded in the unique relational work and ethos of frontline care work contexts. Ultimately, by framing participatory AI around frontline care, our framework presents a structured starting point for reflection around worker-centered design approaches that more comprehensively account for the burgeoning, real-world challenges introduced by AI in frontline care work.

## 2 Related Work

Our paper bridges the CSCW community interested in frontline care (Section 2.1) with the growing community across subfields developing participatory AI (Section 2.2).

### 2.1 Care Systems, Care Orientations, and Care Worker Well-Being at CSCW

Care has been a central concern for CSCW since the advent of health information systems and their role in clinical coordination [46]. This large literature has traced the computerization of healthcare delivery via electronic health records, personal computing, and social platforms, including tools for clinicians, patients, and caregivers—and collaborations between any combination thereof. More recently, care has been not only an important application area for novel collaboration technologies, but also a theoretical lens or orientation [150] for designing relational systems, leveraging a range of different conceptualizations of care as “*service, repair, intimate, burden, infrastructure, and ethic*.” [101]. A growing vein of scholarship draws inspiration from Fisher and Tronto’s feminist care ethic, which frames care as all the labor done to “*maintain, continue, and repair our world*” [45], and casts care systems as a form of infrastructure [21, 139, 145], extending it to consider the sociotechnical assemblages as a whole as we are “*tinkering with bodies, technologies, and knowledge*” [57]. Scholars have drawn on this more expansive aperture to consider networks of care beyond the traditional patient-clinician dyad [99], to look beyond formal care settings towards everyday care in communities [132], and to redesign less-obvious sites of care practice, like remote learning [143] and computer security and privacy [139], towards care infrastructure.

A caring lens on care systems has made central the work of care workers. Care workers have always been notoriously underpaid, overworked, and isolated, as trends in global capitalism have devalued and feminized this essential labor [71, 88, 121, 138]. Amidst the shocks to our collective care infrastructures highlighted by COVID-19, Karusala et al. [69]’s workshop at CSCW 2021 showed the need for the future of care to value care work. Emphasizing care workers’ needs and perspectives is not only a politically important act—it has also provided CSCW with alternative visions for computing research and practice that can inspire work in settings beyond care workers. For example, Ismail et al. [66] investigated how frontline health workers in India navigate digital

systems towards survival, resilience, advocacy, and resistance, and advocated these four principles as care orientations for researchers and practitioners more generally. Karusala et al. [71] investigated tensions between care and scale by exploring care workers' perspectives on automation in chat-based care platforms. Through speculative design with care workers, Karusala et al. [71] found these platforms need not necessarily emphasize “*scale-thinking that promotes the decontextualization of care*”, and could instead emphasize growing relationships and embracing heterogeneity.

In addition to drawing on care workers' perspectives to reimagine collaborative technologies, a growing literature at CSCW has explored how such systems might support the well-being of care workers themselves. Working primarily with knowledge workers, scholars have explored systems for workplace well-being that include how novel sensing might inform managers of workers' stress and productivity, or support self- or collaborative management of occupational health [82]. Much of this work has focused on guardrails for developing such technologies in ways that center workers' autonomy and affirmative consent [24, 28, 29, 72]. Care workers present a novel challenge for workplace well-being technologies in this lineage: care work is more mobile, less artifact-dependent, and highly relational compared to the knowledge work for which many tools are built. Solano-Kamaiko et al. [118] and Adler et al. [2], in studies with home care workers and medical residents respectively, each found that novel well-being technologies like smartwatches and fitness trackers can introduce new data-based windows into FCWs' occupational well-being and productivity, but struggle to capture important aspects of care work: emotional, affective, and articulation work.

Our work builds on this tradition to examine how themes in CSCW's treatment of care and care work extend to the age of artificial intelligence (AI), and the participatory tradition emerging therein. Whether deployed to manage patients' health or to manage care workers' well-being, technologies employing machine learning (ML) and AI are increasingly found across CSCW systems for care, including care robots [147], mobile systems optimizing workflows in care homes [84], and online platforms connecting providers and patients—many of which use automated responses to fill gaps in provider availability [71]. If AI is to be made integral to our care infrastructures, we need to design it in caring ways: e.g., to respect heterogeneity and growing relationships above decontextualization [71], to promote human adaptability rather than AI inflexibility [51], or to foster societal survival, resilience, advocacy, and resistance [66]. To investigate this possibility, we turn to the participatory tradition in technology design.

## 2.2 Participatory AI and Worker-Led Design

The mass release of large language models (LLMs) and their commercialization have brought renewed attention to questions of participation in technology design: how those represented or affected by a given technology can or should be involved in decisions about its development and release. From the datasets they rely on to the compute needed to train them, modern AI is deeply centralized in the hands of large technology companies [80], and even open-source LLMs are effectively closed to third-party audit by their sheer size [144]. That AI is so closed and centralized creates power imbalances between AI developers and the users, consumers, and social worlds their technologies increasingly scaffold.

To counter these imbalances, researchers have proposed applying participatory methods to AI. This effort has been cross-community, inspiring scholars not only at CSCW but also in FAccT, HCI, participatory design, and human-AI interaction. These efforts share the underlying premise that leveraging meaningful stakeholder participation in AI can help communities both benefit from the technology and shape its development—defining its purpose and mitigating potential harms. Within participatory AI, many works make reference to Arnstein's seminal *Ladder of Citizen Participation* [5], a typology of citizen participation in which each rung in the eight-step ladder corresponds to a particular level of citizens' power (e.g., 1 = Manipulation to 8 = Citizen Control).

A meaningfully participatory civic process, Arnstein outlines, moves up the ladder towards citizen control. Horn and Draude [58] translated Arnstein's ladder into a relevant model for HCI researchers working on data-driven applications. Borrowing Mortier et al. [93]'s definition of agency, Horn and Draude [58] describe the goal of a participatory system as one "*concerned with giving people the capacity to act within these data systems, to opt-in or to opt-out, to control, inform and correct data and inferences, and so on*". More recently, Delgado et al. [31] developed the *parameters of participation*, a conceptual framework for how to align participatory goals and methods in AI design, and Suresh et al. [127] outline how to apply participatory methods in the "*foundation model*" paradigm of AI.

While participatory AI continues to explore the potential of community control of AI, debates swirl around the extent to which participatory efforts *can* meet the field's goals. There are methodological incompatibilities: as Young et al. [149] point out, the sheer scale of Big Tech-level modeling efforts strains what we know of tractable participation at the local level. There are also qualities of automated systems that make them difficult to govern through commons: Choi et al. [22] argue AI's tendency towards behaviors unpredictable by system designers means they should also be considered actors in PD engagements [22]. And, as Sloane et al. [115] point out, participation is not a panacea for the harms possible when introducing automated systems to novel settings, and responsible development of AI requires not only creating participatory infrastructures (e.g., community advisory panels), but also leaving room for AI to be discarded as a solution. These debates often resurrect long-held debates in participatory design more broadly. Urging technologists to not forsake the forest for the trees, Bødker and Kyng [16] encouraged researchers to work in partnership with publics on long-term goals, including redistribution of power and resources, rather than artificially narrowing focus on shorter-term, smaller-scale gains—like designed systems [16].

The aforementioned debates in participatory AI resonate with longstanding debates around participatory methods in the lineage of worker-centered design. Increasing workers' participation in governance of their work conditions was the initial impetus for participatory design in the Scandinavian tradition [54]. Working more within HCI, Nardi and Engeström [94] called for designers of information systems to better incorporate the perspectives of the workers who would be using their designed artifacts. As work settings have evolved, so too has the field. Outlining a vision for design to be not only participatory but explicitly worker-centered, Fox et al. [48] advocate for researchers and practitioners to move "*beyond established workplace settings or roles of employment*" and considering the broader social context around the workers and their work.

Taking up Fox et al. [48]'s call, one important strand of recent CSCW scholarship has considered worker participation through the lens of labor rights. Early instances of scientific management such as Taylorism aimed to develop technology to measure worker outputs towards goals of "*efficiency and self-maximization*"—goals defined from the perspective of employers [75]. In contrast, a growing research community has explored how work tracking tools can be leveraged towards advocating for improved work conditions. Prior work in HCI and CSCW present examples of how data collection could help reduce instances of information asymmetry, aid workers in making better decisions, and collect evidence of employer violations of labor law for different kinds of workers, including platform workers [65, 133], gig work drivers [17, 59], and care workers [90, 118].

These recent approaches hold promise for meaningfully participatory technology design in today's work contexts. They center issues like wage theft [86], and create avenues for worker expression through data that are not necessarily mediated through designers' imaginaries—e.g., data collection systems [17, 90]. Still, these approaches carry with them concerns around surveilling workers and imposing unwanted technological burdens [114]. Moreover, there are still questions about how to ensure data-driven approaches are more effective by considering worker perspectives [103], or focusing on mobilizing rather than convincing [27].

Our paper draws on the participatory and worker-led design traditions reviewed here to consider how FCWs can be appropriately involved in building AI for their needs. Frontline care has unique power dynamics between care workers, employers, and care recipients that make questions of meaningful participation more challenging. FCWs are subject to surveillance not only from their employers, as many workers are [24], but also what Mateescu and Ticona [84] describes as refractive surveillance—that of the care recipients (called patients or clients, depending on the care site). FCWs’ efforts at collective action are also often stymied by the self-sacrificial nature of care work—what is often framed as a responsibility to broader well-being [43]. When workers are weighting actions like strikes for better work conditions, they often face a dilemma in “*balancing advocacy and patient well-being*,” specifically “*is it acceptable ...to strike to address issues now so they are able to provide better high-quality care later?*” [110]. Meaningfully participatory technology design of AI, with and for FCWs, must contend with these challenges. We place our work on this frontier.

### 3 Methods

#### 3.1 Combining Theory and Empirics

Our work was inspired by recent scholarship in CSCW and HCI that has drawn together questions from the literature with empirically-grounded observations from researchers’ own experiences—e.g., trauma-informed computing [20], critical race theory for HCI [97], socio-ecological perspectives on vulnerability [128], and social justice-oriented design [36]. Following this tradition, we took a reflexive and autoethnographic approach and added our collective decades of experience as technologists and researchers in a variety of frontline care settings (Section 3.2) to our literature synthesis (Section 2). This approach allowed us to provide concrete examples of our synthesized principles, as seen in Dombrowski et al. [36].

First, we synthesized literature around participation across different disciplines, including discussions around commons, publics, and collective action (e.g., [19, 33, 107]); ladders/levels of participation (e.g., [5, 32, 100]); algorithmic approaches to pluralism and parity (e.g., [61, 67]); consent and refusal (e.g., [24, 63, 151]); sovereignty, stewardship, and governance (e.g., [35, 78, 129]). Then, we highlighted key considerations for participatory AI based on our field experiences (e.g., uses of data for prediction vs. advocacy, types of tracking and data collection that were passive or interpretive, organizational hierarchy and institutional support around data-driven tools) and highlighted specific examples from our research contexts. For this step, we met repeatedly as a research team to share our experiences and iterate on themes inductively, using an approach inspired by grounded theory [18] and thematic analysis [12].

Next, we synthesized these considerations into a framework of guiding questions (Section 4) with which technologists would need to grapple when proposing AI projects (e.g., *What data type, and at what level of fidelity, is needed for the research question? How will you handle conflicts within and between stakeholders? What is the history of introducing data-driven technologies into this ecosystem?*). We organized these questions into a table with more details around our specific research contexts (see Appendix B). Then, to frame our guiding questions, we developed a worksheet following the example of similar conceptual artifacts used in CSCW and HCI (e.g., guidebook for public sector AI [73], framework for explainable AI [40]), in addition to interactive tools used in design research (e.g., canvas for lean UX [53], user persona template [92], Theory of Change worksheet [77]).

We further refined our framework using concrete case studies in frontline care work (Section 5). We filled out the worksheet together based on the case studies we selected to reflect current trends in AI deployment within frontline care settings and vary in both context and proposed AI intervention (see Section 5 for more on case study selection). By applying the worksheet to these real-world examples, we iteratively refined how key concepts were defined (e.g., providing examples to help

distinguish levels of participation), revised the wording and structure of questions (e.g., expanding consideration beyond workers and employers to include additional stakeholders), and adjusted the emphasis of different sections (e.g., increasing space for more extensive responses such as Box 4.1).

### 3.2 Positionality

The authors each have around a decade and a half of experience designing and developing AI systems in frontline care contexts, ranging from home- and community-based care to tertiary clinical care (e.g., uses of software in hospitals), across the U.S. and the Majority World (including South Asia, Sub-Saharan Africa, and Southeast Asia). Across our diverse research contexts, we share a commitment to participatory methods for technology development, as both an object of research inquiry and a methodological toolkit. Within the umbrella of participatory methods, our experiences include: (1) understanding the existing operations and sociotechnical infrastructure within care contexts, through field observations, interviews, and focus group discussions with workers and other stakeholders (e.g., patients, administrators, government officials); (2) exploring potential AI or data-driven systems through design workshops and provocations; and (3) deploying and evaluating designed systems, in the lab and the field. Outside of our work in frontline care, we each also have work experience in the modern technology industry, as software engineers, designers, user researchers, and product managers, across startup and Big Tech corporate environments. This combination of theoretical commitments, on-the-ground research experiences, and real-world work experience led us to the theoretical and translational contributions of this paper.

## 4 Proposed Framework

We now present our conceptual framework offering structured guidance for technologists exploring whether and how to develop participatory AI systems in frontline care work.

*Intended Audience.* We define technologists as people motivated by the challenges in frontline care and the possibility of whether AI can help but are unsure of what exactly to build and how best to collaborate with frontline care workers to make it happen. Whether working in academia, industry, or civil society, technologists may have a remit as engineers, designers, program or product managers, or researchers. They may look to do projects like design explorations, implementations, evaluations, or any combination thereof. Influenced by the tenets of user-centered design and responsible AI, they may want to involve users and communities in the project, and see techniques from participatory design as a starting point for scoping their work. But it remains unclear exactly how to apply participatory AI methods in contexts for frontline care work.

Our goal is to help technologists in this situation define and develop an equitable and participatory project around the prospect of AI in frontline care. Specifically, we guide technologists to consider the critical aspects of an AI project's infrastructure and key considerations for participation in that context. In order to best take advantage of this guidance, we suggest that technologists have experience facilitating co-design using participatory toolkits or related design activities. Moreover, our framework helps prepare technologists for collaboration with FCW partners. However, the technologists themselves are responsible for providing the necessary background to the other stakeholders, including explaining what different participatory mechanisms would entail. In Section 6.2, we describe how future work can help scaffold technologists applying the framework in practice, including trainings and AI literacy initiatives.

Finally, the framework is not intended to be exhaustive nor supplant broader efforts at working with FCWs towards worker justice (we unpack further in 6.1 how our framework relates to these

Participatory AI in Frontline Care Work

Project Description

4.1 AI Intervention

4.1.1 Goal of Applying AI

Problem:

End goal(s):

☐ Understand work/workers

☐ Facilitate or mediate work

☐ Connect workers

☐ Other: \_\_\_\_\_

Existing Approaches & Intended Improvements

4.1.2 Stakeholders

4.1.3 AI vs. Human Trade-Offs

Consider:

• Data representations of constructs

• Comprehensive data vs. surveillance

• Human vs. machine bias

• Efficiency vs. individual burden

4.2 Factors Influencing Participation

4.2.1 Macro

(societal)

Consider:

• Devaluation of care work

• History of research injustice

• Techno-optimism

4.2.2 Meso

(organizational)

Consider:

• Employers, unions

• Care orgs <-> tech companies

• Care orgs <-> regulatory bodies

4.2.3 Micro

(individual)

Consider:

• Financial and time constraints

• AI literacy

• Worker positionality

4.3 Levels of Participation

| Key stakeholders:                                  | Data Collection and Stewardship<br>collecting, processing, storing data | Model Design and Development<br>using data, making technical decisions | System Maintenance and Evaluation<br>upholding performance guarantees, sunset models |
|----------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 0 (inform)<br>one way flow of info                 |                                                                         |                                                                        |                                                                                      |
| 1 (consult)<br>inviting opinions                   |                                                                         |                                                                        |                                                                                      |
| 2 (involve)<br>advise but not decide               |                                                                         |                                                                        |                                                                                      |
| 3 (collaborate)<br>negotiating tradeoffs           |                                                                         |                                                                        |                                                                                      |
| 4 (empower)<br>decision-making or managerial power |                                                                         |                                                                        |                                                                                      |

Fig. 1. Worksheet that can be used to structure conversations about participatory AI systems in frontline care work contexts. See Appendix A.1 for a textual (non-visual) representation of the worksheet questions.

goals). Rather, it offers a starting point for technologists to reflect ahead of discussions with FCW collaborators about the practical and equitable dimensions of participatory AI projects.

*Structure.* The sections of the framework are designed to build on one another. The first section (Section 4.1) guides reflection on whether applying AI in this context is appropriate. If so, the second section (Section 4.2) examines contextual elements that may influence participation. Finally, the third section (Section 4.3) connects these elements to the AI design, development, and maintenance pipeline to help brainstorm appropriate mechanisms of participation.

In this section, we describe the components of our conceptual framework through the framing device of a worksheet (Figure 1) that our intended technologists could use to reflect on applying participatory AI in their specific contexts. Our worksheet draws inspiration from other conceptual artifacts within CSCW and HCI (e.g., Kawakami et al. [73]) and guided activities used in design-thinking workshops by industry professionals to “*expose gaps in the team’s understanding of the problem they’re solving, who they’re solving it for, and why they believe their solutions will work*” (e.g., LeanUX Canvas [53], Theory of Change worksheet [77]). In the prose, we describe how technologists would be guided at each step of our framework to reflect on their proposed projects, and include instructive Box 4.X tags to indicate the corresponding section of the worksheet.

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#### 4.1 Should AI be applied in the context, and how?

Technologists should begin by examining the *appropriateness* of applying AI in their frontline care context, considering whether and how AI *should* be introduced. Weighing the appropriateness of the intervention at the very beginning of the project can help mitigate costly failures before they cause harm to people and communities. Working on applications of AI in public sector decision-making, Kawakami et al. [73] noted that “*many failures in public sector AI projects can be traced back to decisions made during the earliest problem formulation and ideation stages of AI design.*”

This section decomposes the question of appropriateness into a set of guiding probes, each of which we further unpack into specific considerations for frontline care.

**4.1.1 What is the goal of applying AI?** Technologists should begin by identifying the key concerns within the context and exploring how AI can be leveraged to address them. In frontline care work, we identified three potential goals for applying AI: **understanding work or workers** (e.g., analyzing their occupational health and burnout [2]), **facilitating or mediating work** (e.g., conversational agents that answer workers’ questions on-demand [8], robotic assistants that help workers complete tasks [147]); and **connecting workers** (e.g., passing information between workers during handoffs [138], developing solidarity towards collective action [90, 106]). Some AI systems may aim to achieve multiple goals, such as collecting data on workers not only to measure worker productivity but also to use those insights to optimize the distribution of workers across shifts.

Another consideration is whether there are existing technical and/or non-technical approaches to this core objective, and how the introduction of an AI system could improve upon or learn from the current state of practice. For instance, leveraging existing data infrastructures (e.g., time-tracking or work-tracking systems) may help reduce the added burden often placed on workers and ease the transition of adopting new AI systems. Additionally, non-digital processes could still inform the development of digital ones [52]. This is especially relevant in frontline care settings where much of the digital infrastructure may still be relatively nascent. Lastly, existing approaches to workplace technologies and AI have primarily focused on knowledge workers. Assuming workers regularly use digital devices, these approaches have leveraged technological and occupational infrastructure to measure workplace environments (e.g., keylogging to measure productivity [14], warehouse and call center worker tracking [124]). However, FCWs often operate differently—they may move between high- and low-connectivity settings [13], spend variable time in front of screens [138], and perform equal parts emotional and affective labor [89]. These workforce nuances may require AI systems to handle connectivity interruptions, creatively leverage portable or environmental data collection methods, and contextualize social dynamics that are hard to quantify.

In Box 4.1.1, briefly state the problem the AI system is trying to address; the type of end goal(s) the system is trying to achieve; and how the system would improve upon existing technical and nontechnical approaches. This would also be a good place to think about how the unique sociotechnical infrastructure and history of introducing data-driven technologies into this in the context may influence new AI design and development.

**4.1.2 Who are the relevant stakeholders and how will you prioritize them?** It is important to clarify the perspectives of different stakeholders, determine which should be prioritized, and anticipate potential conflicts among them. Workplace technologies are often designed in a top-down manner, and as a result, they frequently fail to reflect workers’ perspectives [48]. In frontline care work, these tools must also navigate complex power dynamics between workers and employers [122]. Identifying and addressing these interpersonal dynamics is especially important, as they can significantly limit workers’ ability to provide meaningful consent [24].

For example, FCWs are often beholden not only to their employers but also to the needs of their clients, a dynamic that may shape how they respond in participatory settings [43]. Specifically, FCWs may feel uncomfortable voicing their own opinions and may instead defer to the wants and needs of their clients (e.g., caring more about the privacy of their clients' information than their own [90]). It is also essential for technologists to engage with existing structures of representational power, such as unions or worker associations, to carefully examine how their interests may align with—or diverge from—those of the workers themselves (e.g., unions can effectively represent worker interests, but may prioritize the collective good over the individual [90]).

In Box 4.1.2, identify the top three to five stakeholders who would interact with the proposed system. For each stakeholder group, specify their key priorities in relation to the problem. Next, diagram any key connections, interactions, perceptions, and potential conflicts among these groups. Where appropriate, group stakeholders together to illustrate how they interact with other groups.

**4.1.3 What are the trade-offs between AI-driven decision-making and human agency in this context?** There may be trade-offs to applying AI to your context compared to other existing human or non-digital approaches. For example, there potentially are questions about **what might be missing from data-based representations**, specifically if the construct is difficult to measure or operationalize (e.g., “humor” as an outcome [11]). Historically, AI and data-driven technologies have inadequately captured emotional and relational aspects of care work [89]. This has allowed those in power to narrowly define which parts of workers' contributions are counted and valued, and has frequently turned care work into data work [116, 119].

On the other hand, there could also be a tradeoff between **more comprehensive data and increased scrutiny**, which could result in surveillance or other privacy concerns. For example, having more in-depth measurements of heart rate and other physiological metrics could give a better picture of worker well-being but could also present approaches that are quite invasive to FCWs' personal well-being [116]. In care settings, this may risk restricting the agency of both care workers and care recipients. For example, tracking FCWs' locations could also monitor and reveal where their clients spend their time [83].

Another tradeoff involves **replacing human bias with machine bias**. Although algorithmic decision-making may appear neutral or objective, such processes can still perpetuate other forms of bias and result in broader negative externalities [81]. Moreover, in care settings, the “human touch,” even if biased, may be critical for resolving conflicts or providing a point person in an otherwise cold and distant process.

Furthermore, new AI systems may create a tension between **organizational efficiency gains and individual operational burdens**. While certain tasks may be completed more efficiently, FCWs' agency can potentially be negatively affected if they must learn to use these systems, audit their outputs, and integrate them into daily workflows—contributing to technostress, or stress linked to technology [141]. In frontline care, this potential added burden is especially salient since FCWs' voices have historically been excluded from the design and development of the technologies they use, often resulting in systems that increase rather than reduce individual burden [138].

In Box 4.1.3, list some of the tradeoffs between a proposed AI-driven system and a more human-driven system in your context. You can use some of the considerations from above as a jumping off point, expounding on how they may be reflected, contradicted, or extended.

## 4.2 What are the specific contextual elements that influence whether and how stakeholders can participate?

After analyzing the goals of the intervention (Section 4.1) to determine the appropriateness of introducing AI to this group of FCWs, technologists should consider how participatory the AI

system can and should be. Prior research (Section 2.2) cautions that simply increasing worker and stakeholder participation is not always ideal, as it can subject workers to potentially exploitative research interactions [103] that may place them in the position of solving problems imposed by broader structural forces [55]. Therefore, careful consideration needs to be made about how to mitigate harms that FCWs and other stakeholders might face in participation, including how to lower barriers to entry or reduce risks of retaliation.

We split this question into two interlocking parts: understanding the macro, meso, and micro factors that influence stakeholders' capacity and willingness to engage in a participatory AI system (this section); and understanding where within the participatory AI pipeline FCWs' participation is therefore best enlisted (Section 4.3).

**4.2.1 What are macro factors influencing FCWs' participation in AI development?** To work with FCWs, technologists should first understand that **care work has been systematically devalued in contemporary society**, making FCWs often hesitant to participate in technological research and development. Historically, care work has been considered "reproductive labor" rather than "productive labor" towards societal goals [44], done mainly as unpaid domestic work in a "gendered and de commodified space, away from the market" [84]. Even as care work is becoming increasingly recognized as "essential" through the pandemic and increased rates of aging [130], technological systems that are built to facilitate and support care often end up further invisibilizing key contributions FCWs make by not counting them or unknowingly adding to them [89].

FCWs also often occupy positions of lesser economic and social status, which may impact how organizations perceive and value participants' input [62]. Technologists should acknowledge the **history of research injustice**, particularly in the U.S., where relationships between researchers and vulnerable populations have long been strained. This legacy includes many of the communities that FCWs come from and work in. When technological development projects do not acknowledge and work against this injustice, they can create environments not for collaborative generation but instead what Harrington et al. [55] call "*blue-sky ideation*" or place undue "*epistemic burden*" [103], which can ultimately feel like a waste of participants' and communities' time. Moreover, this history may also impact whether FCWs feel like their voices will be heard or how they are expected to respond, leading to potential response bias [87].

Trends towards devaluation of care work and workers make FCWs particularly susceptible to **techno-optimism around AI**. Similar to examples of other response biases among participants who might be susceptible to "novelty bias" about ideas presented by researchers [87], research has shown FCWs can ascribe unrealistic expectations to these technologies, or favor AI systems' judgments over humans' [116]. As technology companies and investors push consumer-grade AI applications into mass markets, popular narratives of superhuman AI can make participation in technology design daunting or fraught for the people who work one of our most human jobs.

In Box 4.2.1, list some of the applicable macro or societal level factors. Consider some of the historical forces around power hierarchies in frontline care work and also your specific contexts.

**4.2.2 What are the meso factors influencing participation?** Beyond societal-level phenomena, technologists should consider how FCWs' participation is shaped by the groups and organizations they inhabit, and how those organizations interact. For example, FCWs are often **employed by larger organizations**, such as agencies, which may operate under for-profit or non-profit models, and have highly top-down, directive structures. Additionally, some workers may belong to **labor unions or other collective action entities**, which likely foster bottom-up, empowerment-first cultures that encourage greater collaboration and participation. Worker cooperatives are also being

piloted in some frontline care contexts, in which workers are both owners of the operation and representatives of their own interests [104].

Employers often do not develop technologies in-house, but rely on external **technology vendors**. Vendors are responsible for making many of the design decisions that result in the technical infrastructures provided to workers. However, they rarely have firsthand experience with care work, and often have few opportunities to learn from workers' lived experiences. Employers are usually responsible for representing worker interests to vendors during technology development—but employers may themselves have little insight into worker realities [85] and how new technologies could impact FCWs' occupational well-being, as seen in the home care industry [117].

Another often-relevant relationship is that of the care organization and the **regulatory bodies** that influence how FCWs are compensated (i.e., through what can be insured) and, in turn, what technology is used. Whether in highly regulated contexts like health or less-regulated contexts like consumer products, there may be standards around how information can be stored and processed, which could result in systems made of disconnected records spread across different providers and databases [113]. Moreover, there may be laws that dictate which types of technology have to be used, including for monitoring work to ensure these payments are not being taken advantage of [83]. This arrangement often dictates employers' decision-making around digital tools, leading them to prioritize those aimed at employee compliance and insurance billing rather than the needs and well-being of FCWs. These economic pressures can significantly affect employers' willingness to engage participants authentically, the goals of AI tools, and the capacity for meaningful change.

In Box 4.2.2, list some of the relevant meso or organizational level factors for your context. Consider what some of the key institutional players are at this level that may influence what technologies are being used and how they interact with each other.

**4.2.3 What are the micro factors influencing participation?** The macro and meso factors described above can manifest at the micro level, in individual FCWs' experiences and capabilities around participation. Technologists should consider first and foremost that FCWs are often **resource-limited**, meaning they can also face significant constraints on the time, money, and energy they can pour into technology design. These resource limitations can be addressed with compensation for research participation, e.g., by offering a participant a gift card in the amount of their hourly rate for an hour spent on a research study. However, technologists should still be cognizant of additional costs borne by participants: transportation costs, childcare costs, the costs of switching their attention from their work to participating in research, and other affective and material burdens of contributing their stories to research [38].

FCWs can also have **low AI literacy**, which can limit their ability to meaningfully participate in AI design and governance. Many FCWs do not receive training in computer science, software engineering, statistics, or machine learning, and have scant resources for learning on-the-job—these types of skills may ultimately not be core to providing care and can actually distract from it. However, some level of AI understanding is necessary for participation. Existing frameworks often assume participation in AI takes significant computational and mathematical skill (e.g., understanding probability or data structures/algorithms). However, it is often necessary to narrow down which types of skills are crucial to develop a critical eye towards what the technology can and cannot do for participation and develop unique ways to nurture this in their participants.

Lastly, FCWs' individual experiences can be influenced by **a tendency to place their clients' and employers' needs above their own**. Inherent to care work is a spirit of self-sacrifice; across care contexts, we see FCWs defer to their clients' needs and wishes when it comes to data privacy [90, 136], what they compensated for their work [47], and even their own personal health [121]. Participatory design can be an uncomfortable and foreign process to many workers who

are accustomed to working “*in the background*”, often going unacknowledged and performing “*invisible*” labor [88, 89]. They may not be used to having their perspectives centered by others, especially those in positions of power, within the broader care ecosystem.

In Box 4.2.3, list some of the applicable micro or individual level factors for your context.

#### 4.3 For each stage of the participatory AI pipeline, what is the appropriate level of participation?

Having understood the macro, meso, and micro factors that may influence FCWs’ capacity and willingness to participate (4.2), technologists are then equipped to analyze where along the participatory AI pipeline FCWs are best engaged. We envision this analysis as a mapping of each pipeline stage to the appropriate mechanisms for participation.

We consider the participatory AI pipeline to consist of three stages: (1) data collection and stewardship; (2) model design and development; and (3) system maintenance and evaluation. We designed this framework based on existing literature (cf. [25, 42, 60, 112, 126, 127]) and our experiences of machine learning pipelines in real-world frontline care work. Scholars have described AI pipelines in many different forms; for the present study, we synthesized a tractable three-part conceptualization, focusing on phases that have the most distinct considerations for participation.

Some projects start by assessing the data available to represent a certain phenomena (e.g., worker productivity). Storing, curating, pre-processing, profiling, and protecting this data is a distinct stage of work that often employs archival methods and may be completely upstream from any use of said data in machine learning pipelines; hence we describe it as **data collection and stewardship** (stage 1). In the case where a given dataset is selected and ingested for training, validation, and/or testing of a model, these activities constitute their own stage, **model design and deployment** (stage 2). This stage also includes making and documenting decisions around technical architectures and performance guarantees, including when and how to incorporate explainability tooling. Finally, we consider the post-deployment maintenance of a model to be its own stage: **system maintenance and evaluation** (stage 3). The labor of maintaining a model and a working system long-term is crucial to upholding its performance guarantees, especially as a check against the distribution drift that might be natural over time. This stage also includes the critical yet often thankless coordination work of organizing stakeholders to carry out the maintenance (e.g., how often the model should be checked, and by who), and the ability to sunset models that no longer perform well enough for deployment. In contemporary generative AI pipelines, this last stage also encompasses post-training efforts at safety testing (e.g., AI red-teaming), and filters and safeguards on user prompts.

For each stage, we consider the possibility of participation on a 5-tier spectrum adapted from Patel [100]’s modes of participation. Each tier contains specific mechanisms for participation, collated from Suresh et al. [127] and Tseng et al. [136], and the tiers are listed in ascending order of their participatory quality. Tier 0, **inform**, is for use when the burdens of participation (understood via the macro/meso/micro mapping in Section 4.2) are too great to justify the benefits and the decisions are made by technologists and communicated to other stakeholders in a one way flow of information. Adding more participatory mechanisms to tier 0 would lead to tier 1, **consult**, in which the FCWs at hand face fewer but still significant burdens to participation and are asked to provide unidirectional opinions on created artifacts (e.g., via mechanisms like surveys, interviews, or A/B testing). More participatory than tier 1 is tier 2, **involve**, where FCWs might be asked to deliberate design decisions and created artifacts, via mechanisms like citizen juries or public dialogues, but ultimately are not making the decisions. In situations where FCWs have more capacity and willingness to participate, technologists might consider mechanisms in tier 3, **collaborate**, where FCWs co-design systems that split the burden of effort between participants and researchers (e.g.,

data trusts or data donation). FCWs are invited to negotiate tradeoffs of different decisions and with assurances that technologists will try their best to incorporate advice and recommendations. Finally, there may be cases where FCWs are best positioned for the **empower** pipeline step, meaning they would be responsible for the full scope of making decisions related to the given stage. These cases might include communities that already have established practices for archiving and managing their data, or in-house resources for model development and evaluation.

To complete the table in Box 4.3:

- (1) In the box labeled “Key stakeholders,” list out the two to three key stakeholders that could be participating in each part of the process (e.g., FCWs, their employers, their clients).
- (2) In the first row of the table, fill out in broad strokes what each stage of the project could look like (i.e., the types of data inputs/outputs, potential functions of AI, processes for maintenance/evaluation).
- (3) In each of the internal cells of the table, fill out examples of participatory mechanisms that would make sense for your context. You may decide to use the same mechanism across different stages or skip filling out some of the boxes where the participation does not necessarily apply. You may also decide to separately indicate different mechanisms of participation for different stakeholders in cases where the methods may not align.
- (4) Select the corresponding mechanism(s) that might make the most sense for your project given the identified macro/meso/micro factors. You may decide to use or provide the option for multiple mechanisms in a given stage.

## 5 Case Studies

To illustrate the application of this framework, we provide completed worksheets for two contrasting case studies (visual versions: Figures 2 and 3; text-only versions: Appendices A.2 and A.3). The following sections are constructed to showcase the content of the worksheets for each of the case studies. Firstly, we include background information, including considerations based on questions in Box 4.1 about the goals, stakeholders, and tradeoffs of using AI in this specific context. Then, we also present several tensions that arose in the worksheets about both the Box 4.2 factors influencing participation and Box 4.3 potential levels and mechanisms of participation.

We selected these specific case studies from some recent and emerging trends in AI and frontline care to measure, monitor, and surveil workers, such as ShiftMatch.AI [1]. We used a set of criteria to make sure these case studies highlighted key areas of contrast across the design space in frontline care work: (1) the context and (2) the AI or data intervention proposed. With respect to (1), we selected case studies that contrasted across key areas of difference such as the FCWs in question and their anticipated familiarity with technology (e.g., low-wage workers vs. physician trainees), and the type of workplace at hand (e.g., home-based vs. hospital-based). For (2), we considered contrast points in the case studies like which aspect of labor the system is meant to enhance (i.e., making workers more efficient vs. taking care of their well-being), what types of goals the system could achieve (i.e., optimizing outputs vs. uncovering new patterns), and the predicted time horizon for the system (i.e., adapting existing algorithms v.s. aiming for moonshot innovations in AI).

### 5.1 Case Study 1: Shift Matching in Home Care (Figure 2 or Appendix A.2)

**5.1.1 The Context of the Proposed AI.** The first example explores how AI could be used to distribute shifts for home care workers (HCWs). HCWs are paid professionals who care for older adults and people with disabilities in their homes. They are often women, people of color, and/or immigrants who face financial precarity and an emotionally and physically challenging workplace [102]. HCWs often work for agencies in a distributed and shift-based manner: a central administrator

## Participatory AI in Frontline Care Work

Project Description *AI to allocate shifts for home care workers*

| 4.1 AI Intervention                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>4.1.1 Goal of Applying AI</b><br>Problem:<br><i>shift matching takes a lot of time and energy from home care agency coordinators</i><br>End goal(s):<br><input type="checkbox"/> Understand work/workers<br><input checked="" type="checkbox"/> Facilitate or mediate work<br><input type="checkbox"/> Connect workers<br><input type="checkbox"/> Other: _____<br>Existing Approaches & Intended Improvements<br><i>currently matches are done by hand by admins based on their knowledge of the worker/patient schedules, preferences, needs, skills</i><br><i>AI could be more efficient / improve care</i> | <b>4.1.2 Stakeholders</b><br> | <b>4.1.3 AI vs. Human Trade-Offs</b><br>Consider:<br>• Data representations of constructs<br>• Comprehensive data vs. surveillance<br>• Human vs. machine bias<br>• Efficiency vs. individual burden<br><i>miss contextual knowledge / rapport (personality/style, empathy/compassion)</i><br><i>already isolated workers may lose the opportunity to check-in with a human</i><br><i>first-come-first-serve matches may bias towards "more online" workers, location biases, etc.</i><br><i>burden on workers to identify, escalate, and fix potential issues</i> |

| 4.2 Factors Influencing Participation                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>4.2.1 Macro</b> (societal)<br>Consider:<br>• Devaluation of care work<br>• History of research injustice<br>• Techno-optimism<br><i>bias towards quantifiable factors may further invisibilize emotional/relational contributions/expertise of workers</i><br><i>minoritized workers can again have their data / participation misused</i> | <b>4.2.2 Meso</b> (organizational)<br>Consider:<br>• Employers, unions<br>• Care orgs <-> tech companies<br>• Care orgs <-> insurance<br><i>existing union presence a way for workers' voices to be collected / heard</i><br><i>home care agencies feel less confident disputing narratives from tech companies</i><br><i>employers may be incentivized to retaliate against workers</i> |

| 4.3 Levels of Participation                                                             |                                                                                                                                           |                                                                                                           |                                                                                                         |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Key stakeholders:                                                                       | Data Collection and Stewardship                                                                                                           | Model Design and Development                                                                              | System Maintenance and Evaluation                                                                       |
| • tech companies that are proposing this software<br>• employers<br>• home care workers | collecting, processing, storing data<br>existing/future matches + prefs/satisf/needs/skills                                               | using data, making technical decisions<br>agency criteria + worker/patient data -> ranked list of matches | upholding performance guarantees, sunset models<br>periodic assessments of efficacy + model fine tuning |
| <b>0 (inform)</b><br>one way flow of info                                               | employers are instructed (but don't give feedback) on how matches + data are collected                                                    | model is developed/deployed on static datasets, no input beyond that                                      | employers are informed about the results of periodic assessments                                        |
| <b>1 (consult)</b><br>inviting opinions                                                 | interviews with workers/employers to understand data collected                                                                            | employers and workers give feedback on the matches thru interviews + surveys                              | survey for employers / workers to report issues                                                         |
| <b>2 (involve)</b><br>advise but not decide                                             | advisory board of workers and employers that can share perspectives on data collection + how to operationalize data                       |                                                                                                           | workers and employers upvote/downvote on which issues they want addressed                               |
| <b>3 (collaborate)</b><br>negotiating tradeoffs                                         | workers select stewards (union, other workers, coordinators) to be involved in conversations about how to manage their data and its usage |                                                                                                           | stewards periodically reassess system performance and update the design                                 |
| <b>4 (empower)</b><br>decision-making or managerial power                               | power to decide what data is collected, veto usage, and what data is used for                                                             | cooperative ownership of the data used by the system and the model                                        | total control over defining and addressing model performance                                            |

Fig. 2. Case Study 1: Example of how the worksheet could be completed regarding the case study of using AI for shift matching in home care. The completed worksheet raises key considerations around handling relational elements of care and the participation of overburdened and low-wage workers. See Appendix A.2 for a textual representation of the content in the completed worksheet.

dispatches them to different clients' homes, and they may hand off to different HCWs between shifts. This relationship between workers, clients, and employers is depicted in Box 4.1.2 of Figure 2 or Appendix A.2, which details the long-term nature of the relationships between the different stakeholders, and how these relationships influence how they perceive each other.

In the U.S., state and federal regulations require home care agencies to log when care visits occur, and what tasks are completed during a visit, in order to optimize health insurance payments for home care services [96]. Many agencies thus require these FCWs to use telephone-based systems to log their tasks—systems that are often buggy, cannot be easily corrected, and burdensome to use, but nevertheless determine when and how home care workers get paid [84, 98]. Recently, CareConnect, a vendor of a widely used telephone-based work-tracking system, proposed “ShiftMatch.AI,” a system leveraging work-tracking data to help the home care agencies more effectively match care workers and clients, to “drive revenue, reduce costs, improve satisfaction, and seamlessly scale” [1]. This proposed use of AI attempts to “facilitate or mediate work”, by addressing the difficulty and inaccuracy of shift matching by human administrators (Box 4.1.1).

Working through the human-AI tradeoffs involved in this proposed intervention (Box 4.1.3), we find several challenges that need to be carefully considered. Technologies previously deployed in

the home care space can end up surveilling both care workers and care recipients [83], invisibilizing the emotional and relational contributions of the workers [89], and creating additional technological burden for the workers, including the work to find and fix issues [138]. Thus, technologists need to consider the tradeoffs so that new AI systems would not exacerbate these dynamics for the essential but vulnerable workers [116]. AI systems would want to build on the personal rapport that human administrators had with the workers that provide much needed person-to-person contact for alienated workers spread out across client homes. At the same time, they would want to ensure that they do not perpetuate human bias and maintain a process that is amenable to triage and audit. Moreover, new AI systems need to ensure that the burdens for identifying, handling, and escalating issues would not fall upon the shoulders of workers who are already overburdened with other forms of data work [125] or the maintenance of the system more broadly [142].

**5.1.2 Challenges and Opportunities for Participatory AI.** Applying our framework raises a key challenge for AI in frontline care work: ensuring that AI-driven systems account for the less tangible but still important relational factors in care, such as the personality match between workers and clients, or more nuanced understanding of the workers' schedule outside of work. This concern is culminated in the macro factor influencing participation in Box 4.2.1 of Figure 2 or Appendix A.2, where the historical bias towards quantifiable factors that could impact what is being optimized for by AI systems. Therefore, future AI systems and "care algorithms" [147] would want to take into account the emotional and relational contributions and expertise of care workers.

This case study also raises considerations about working with workers like home care workers, who often face many burdens. As noted in Box 4.2.3, on a micro or individual level, increasing participation of workers in the design and deployment of AI systems may place responsibilities on workers that they are not being compensated or trained for and that further complicate their schedules that are at the whim of their employers or driven by their clients' needs. FCWs may also fear retaliation from their employers, which means on the meso or organizational level (Box 4.2.2), employers also need to be accountable for their actions, so as not to simply repeat unjust research practices present on a societal level (Box 4.2.2). Moreover, they need to be aware of their own AI competency and literacy and potential tendencies towards techno-optimism.

Therefore, the participatory mechanisms selected in Box 4.3 would need to consider the resources FCWs might need to participate; ensure adequate sociotechnical infrastructure and AI literacy; and mitigate the potential risks of participation. We selected a participatory approach of stewardship for the data and model parts of the AI pipeline inspired by Patel [100] and Tseng et al. [137], where FCWs select representatives to be involved in conversations about how to manage their data and its usage, echoing existing structures for collective bargaining supported by the unions. Moreover, we also selected an option for FCWs and employers to give feedback on the system, but where technologists would still hold the main responsibility for prioritizing and fixing issues that arise.

## 5.2 Case Study 2: Burnout for Resident Physicians (Figure 3 or Appendix A.3)

**5.2.1 The Context of the Proposed AI.** The second example explores how AI could be used to identify burnout among medical residents from multimodal, ambient data (e.g., inferring a resident's work stress from their step counts, note-taking practices, sociality, and other trackable behaviors). U.S. medical residency is a grueling training period in which physicians-to-be ("residents") work long and unpredictable shifts on-call at the hospital. [6]. Developed as a form of apprenticeship, residency has become notorious for negatively impacting well-being: the rate of depression among medical residents (25-33%) is twice the rate of other young adults (8-15%) [2]. The impacts of residency on the medical workforce have been so concerning that in 2017, the governing body of American

## Participatory AI in Frontline Care Work

Project Description

AI to identify burnout among medical residents based on multimodal data

### 4.1 AI Intervention

#### 4.1.1 Goal of Applying AI

Problem:  
*medical residents are burned out and its hard to identify + intervene systematically*

End goal(s):

- ☒ Understand work/workers
- ☐ Facilitate or mediate work
- ☐ Connect workers
- ☐ Other: \_\_\_\_\_

Existing Approaches & Intended Improvements  
*other contexts have used approaches like surveys + physiological measures*

*there is also research exploring how burnout can be identified from clinical notes*

#### 4.1.2 Stakeholders



#### 4.1.3 AI vs. Human Trade-Offs

Consider:

- Data representations of constructs
- Comprehensive data vs. surveillance
- Human vs. machine bias
- Efficiency vs. individual burden

*residents may exhibit response bias (i.e., act differently because they are being observed) and skew the data*

*regulations (e.g., HIPAA) may restrict the type of data that could be used*  
*difficult to balance figuring out who to target for accountability vs. retaliation*

### 4.2 Factors Influencing Participation

#### 4.2.1 Macro

(societal)

- Consider:
- Devaluation of care work
  - History of research injustice
  - Techno-optimism

*qualitative / relational data / contributions may still be undervalued in comparison to more quantitative measures*

#### 4.2.2 Meso

(organizational)

- Consider:
- Employers, unions
  - Care orgs <-> tech companies
  - Care orgs <-> insurance
- insurance companies may feel the original purpose of clinical notes is diluted if the data is used in this new way*  
*representational groups (e.g., ACGME) may not have insight into individual workplaces*

#### 4.2.3 Micro

(individual)

- Consider:
- Financial and time constraints
  - AI literacy
  - Worker positionality
- difficult to engage workers already at high risk of burnout*  
*getting baseline measurements for burnout requires continuous participation from residents*

### 4.3 Levels of Participation

Key stakeholders:

- researchers / technologists
- exploring this approach
- supervisors / hospital admins
- residents

**Data Collection and Stewardship**

collecting, processing, storing data  
*building on existing clinical note taking + feedback process*

**Model Design and Development**

using data, making technical decisions  
*training AI on diff sources of data to identify burnout*

**System Maintenance and Evaluation**

upholding performance guarantees, sunset models following HIPAA (etc)  
*guidelines on how frequently to review/update system*

0 (inform)

one way flow of info

*the system is designed, developed, and evaluated by the researchers / technologists and then residents / supervisors are instructed on how to use the systems*

1 (consult)

inviting opinions

*residents/supervisors give feedback on the kinds of data that could be used*

*residents are asked to rate the accuracy / appropriateness of potential outputs of the model*

*residents/supervisors have the option to fill out a feedback form if issues arise*

2 (involve)

advise but not decide

*residents develop paradigms around which data is used that are available to the system*

*residents/supervisors are brought together to discuss inputs/outputs to the model*

*focus groups are held periodically for feedback from residents/supervisors*

3 (collaborate)

negotiating tradeoffs

*residents can set rules around which kinds of data are "donated" to the model*

*residents debate pros/cons of potential interpretations of the data from the model*

*residents/supervisors help prioritize system issues*

4 (empower)

decision-making or managerial power

*residents have the ability to run the analysis process on their data when they want to*

*residents can personalize how the model works for their own situations*

*resident+supervisor council is responsible for ensuring the system continues to work*

Fig. 3. Case Study 2: Example of how the worksheet could be completed regarding the case study of using AI to infer and intervene on burnout based on multimodal data from resident physicians. The completed worksheet raises concerns around reconciling variable data across individuals and broader accountability infrastructures within the clinical setting. See Appendix A.3 for a textual representation of the content in the completed worksheet.

medical residencies, the Accreditation Council for Graduate Medical Education (ACGME), issued a new mandate for programs to prioritize resident well-being [15]. The relationships between residents, their supervisors, and the hospital admins are in Box 4.1.2 of Figure 3 or Appendix A.3.

Inspired by this mandate, researchers and residency program directors across the U.S. began exploring programmatic and technical interventions to reduce burnout. These include approaches from the subfield of ubiquitous computing, a subfield which seeks to use data streams from technologies already embedded within clinical workflows (e.g., clinical notes and action logs [79], medical billing data [10], physiological data like step counts and activity [2]) to measure and predict when a resident might be experiencing work stress. Within this overall vision, there are several steps where ML and AI could be useful, including inferring work stress and burnout from heterogenous, messy, and longitudinal data sources and combining them to recover more latent signals (e.g., joining information from a resident's clinical notes and their step counts). And any algorithm development in this direction requires technologists to carefully consider the reliability and validity of their work, especially for underrepresented populations [3]. Beyond measurement of potential burnout, AI can also be used in intervention. If reliable and valid algorithms for burnout inference were established, they could be embedded into an overall burnout prevention system personalized for

an individual resident's stressors and preferred care patterns, including self- and collaborative management of work stress, and perhaps even human-AI tools like conversational agents. In our exercise, we described these potential uses of AI as "understanding work/workers" (Box 4.1.1).

Using AI for burnout prevention does, however, require careful navigation of tradeoffs between human and machine agency. In Box 4.1.3, we focused our analysis on three tradeoffs in particular. First, on a basic scientific level, inferring burnout from data traces may fail if workers are made aware of which traces are used to predict their work stress, and change their behavior accordingly. This phenomenon, known as response bias, would skew the data used for the resulting inference algorithms. Second, there remain questions around which data streams are appropriate for use. The regulatory environment around medical residency may restrict which data streams can be incorporated into burnout management, especially patient-related data, or what workers would consider private communications. Finally, any AI-based intervention would need to be embedded in a broader programmatic infrastructure that carefully balances algorithmic support and punishment: e.g., workers should be assisted and not unfairly penalized when the algorithm detects they are experiencing higher work stress. Addressing each of these tradeoffs requires participatory mechanisms—to appropriately involve residents in design decisions, and ensure they are part of the maintenance of the resulting system going forward.

*5.2.2 Challenges and Opportunities for Participatory AI.* This case study highlights the importance of flexible approaches to participation, especially for applications of AI that involve heterogeneous, messy, multimodal, and repurposed data streams. Integrating multiple modalities of data, for example quantitative and passively sensed activity measures with qualitative work stress diaries, might more comprehensively represent the construct of burnout than one modality alone. But at a micro level, questions remain around FCWs' willingness to provide detailed qualitative data—doing so is burdensome, and these workers are by definition already stressed (Box 4.2.3 in Figure 3 or Appendix A.3). Erring on the side of quantitative and easy-to-implement data streams might, however, inadvertently exacerbate macro trends in how qualitative and relational contributions are less-valued than quantitative ones (Box 4.2.1). Still, studies repeatedly show workers want the ability to give context to raw measures like their activity counts [2]—suggesting the participation mechanisms offered to these workers need to be unburdensome yet nuanced.

To complicate the picture, a resident's participation in the system needs to be continuous yet privacy-preserving. Because burnout is a sensitive and highly personalized construct, longitudinal data on a resident's work activities is needed to establish their individual baselines, and detect when they deviate from their regular work stress. This type of participation can be uniquely burdensome: logging work stress over weeks or months can also create work stress (Box 4.2.3). There can also be challenges in how to make the data actionable. Medical residents are relatively privileged compared to FCWs like home care workers; however, they still operate within the hierarchies of hospital settings. On a meso level (Box 4.2.2), this means considering which stakeholders have the capacity and authority to make changes to residents' situation, were they to receive measures of residents' burnout from the proposed system. For example, working within one U.S. medical residency program, Adler et al. [2] found that attending physicians who supervise residents often did not want to receive steady measures of their work stress, because they did not feel empowered to make the necessary changes and did not want a pseudo-clinical relationship with their mentees.

From our analysis of factors influencing stakeholders' potential participation (Box 4.2), we conclude the participatory mechanisms that might work best for residents' burnout need to remain open to different types of inputs from FCWs and their supervisors, including asking for their feedback on the proposed data to use, or having more in-depth discussions about how the model could be constructed and how the system could be maintained. This would help make the AI

system more flexible across different types of situations FCWs may face; build on already strong relationships between FCWs and other stakeholders; and still bring to the table those who would be ultimately responsible for making programmatic changes. Note that in selecting participatory mechanisms at the levels of 1 and 2, we are choosing conservatively, to minimize the burdens that the system would levy upon already-stressed residents. Participatory mechanisms at the levels of 3 and 4 would be ideal in an environment where residents have the time and energy to devote to running their own burnout mitigation system—and in some programs where residents have more free time, this may be possible. In choosing 1 and 2, we err on the side of the minimal participation acceptable, to make room for programs in which residents do not realistically have that agency.

## 6 Discussion

This paper presents a conceptual framework that synthesizes literature on participatory design, frontline care work, and AI, alongside our extensive experience designing and developing AI systems. The framework aims to address three main areas: assessing whether an AI intervention is appropriate for a given project context (Section 4.1); examining the macro (i.e., societal), meso (i.e., organizational), and micro (i.e., individual) factors that shape appropriate participation (Section 4.2); and analyzing these contextual factors to understand which specific mechanisms of participation might apply (Section 4.3). We present this framework as provocation, as a set of guiding questions for technologists, visualized also as a worksheet (Figure 1 or Appendix A.1) that can be used to inspire reflection.

To further operationalize our efforts and examine key tensions that emerge through our framework, we provide two illustrative examples from contrasting frontline care work case studies and apply the worksheet for each case study (Section 5). We consider the challenges of engaging with FCWs for whom technology is often a peripheral component of their jobs and how top-down technological systems risk devaluing their emotional labor while introducing additional operational burdens on workers. We also analyze how AI systems might engage with care as a relational and indeterminate concept—such as facilitating personality matches or processing heterogeneous data from a wide range of care workers. Finally, we reflect on how these systems could contribute towards improving care infrastructures if they strategically involve key stakeholders and amplify the voices of those most affected within these intensely hierarchical settings.

In the following section, we unpack how discussions raised by this framework could shape future research agendas in CSCW. We then close by describing the key limitations of our work, and how future areas of research can expand our framework's practical application and breadth.

### 6.1 Implications for CSCW

*Participation and Privacy in AI-Mediated Frontline Care Work.* First, our framework motivates research challenges for CSCW at the intersection of participatory AI and privacy, where frontline care work can be a keystone case study. Following scholarship on surveillance in care and frontline work (e.g., Mateescu and Ticona [84], Sum et al. [124]), our framework encourages technologists to acknowledge that the data produced in frontline care tracks the movements and behaviors of FCWs and care recipients alike, and may simultaneously serve as health information, productivity metrics, occupational hazard reports, or research data. Thus, our framework underscores the need to overhaul the regulatory standards governing the secure handling of multipurpose data in these socially complex contexts, and design standards that update our notions of secure, private, and trustworthy AI for conditions of refractive surveillance (cf. [131]). Potentially useful here are ongoing discussions within participatory AI of how individuals and collectives can exert control over data [107], including approaches such as data cooperatives, data stewardship, and ultimately, data solidarity [108]. To complicate matters, in frontline care work, the challenge of private data

infrastructure is amplified because frontline care settings are workplaces. It is yet unclear how participatory privacy mechanisms like dynamic consent or data solidarity [91] can be effective in dynamic and often unstable work environments, such as the high employee turnover and variable scheduling seen in home care contexts [86]. To investigate these tensions, CSCW scholars might collaborate with researchers in computer security and privacy to develop frameworks and techniques for privacy preservation in AI that account for frontline care as a context for work (e.g. Adler et al. [3], Chouhan et al. [23]).

*Preserving the Human in Frontline Care.* Our framework also deepens CSCW's understanding of frontline care work as a distinct domain of study, particularly in relation to AI and participation. For instance, the occupational pressures toward self-sacrifice (e.g., FCWs often prioritize patients' needs over their own) and the power asymmetries shaped by intersectional worker identities (e.g., FCWs are often women of color or immigrants, working long hours in isolation for low pay) set frontline care apart from other domains of community-engaged design. The intensely emotional and relational nature of care work further complicates these dynamics, often rendering traditional approaches to participatory AI insufficient or ill-suited.

This raises important questions about how AI can be meaningfully adapted to accommodate these unique characteristics, and what effective implementation looks like in such settings. It also prompts broader reflection on the relationship between care and AI. The growing attention on AI in care may, in turn, help make care work itself more visible—providing technologists and researchers with more opportunities to learn from FCWs across diverse settings. This short-term awareness could be embedded into the long-term design of future systems. It could also present opportunities to collect data on both the concerns and contributions of FCWs in ways that could inform policy [90]. More broadly, it could help elevate care work as both valuable and fundamentally irreplaceable—as the “last bastion of ‘human’ work” [84]. In doing so, it may also highlight the essential human-to-human dimensions of other professions in an era increasingly shaped by automation and AI. A deeper understanding of what constitutes essentially “human” work could help technologists make more informed decisions about which aspects of sociotechnical systems to automate and which to assign to the human-in-the-loop in the most effective and responsible ways.

*Worker Voice and Justice.* Technologists motivated by the problems in frontline care work often face an essential tension: how to amplify worker voice in technology design and move towards justice while balancing the potential harms and benefits of worker participation. Our framework offers technologists structured ways to think through these tensions, grounded in the literature on participation and the practical realities of engaging in the participatory AI pipeline.

Our work opens avenues in CSCW for further research on how to amplify worker voice in technology design by making participation more tractable and meaningful. Scholars can explore creative strategies to reduce the costs and enhance the benefits of participation for individual workers, such as compensating community residents as co-designers or bartering time in exchange for access to resources [55]. Additionally, benefits could be extended to collectives, by fostering connections among isolated workers [105] or leveraging other stakeholders through mechanisms like data intermediaries [58]. Ultimately, this line of inquiry could help reimagine longer-term and more reciprocal relationships between communities and technologists.

We also note that some of these costs of voice may not necessarily be resolved simply through more creative mechanisms of participation. We emphasize the importance of attending to the experiences of low-wage, minoritized workers, who may face research injustice if their participation is approached in superficial or extractive ways. This must also be balanced against the risk of interpreting participatory outputs in ways that place undue responsibility on individual workers to solve structural problems—particularly given that many are already performing “glue work” to

keep their jobs and lives functioning smoothly [4]. Ultimately, as literature in citizen participation [62] and participatory design [16] observe, participation may not always be a panacea.

Importantly, we note that simply amplifying worker voice does not necessarily equal justice. We do not claim our framework should supplant larger efforts to support workers, such as labor organizing. Critiques of ethics washing in AI have noted how toolkit or checklist approaches to responsible technology development can flatten context and sidestep organizational power dynamics [146], neutering justice efforts rather than supporting them. Mindful of these critiques, we offer our framework as a set of guiding questions rather than prescriptions. For technologists in positions to decide what to research or develop, we hope our work can help scaffold context-specific and power-aware deliberation on what they can do within their own capacities for change.

## 6.2 Limitations and Future Work

This paper proposes a conceptual framework grounded in existing literature and our own experience, similar to contributions in HCI and CSCW that provide directions for future researchers, like social justice-oriented interaction design [36], postcolonial computing [64], and feminist HCI [7]. Like these inspirations, our contribution is empirically-grounded in our own expertise and experience as technology researchers and our focus on FCWs, rather than empirically validated. There is ample room for colleagues in CSCW to extend and refine the framework based in their own contexts.

*Enacting the Framework.* We see particular promise in future work that explores practical application of the framework through participatory design workshops [73, 123] and develops the facilitation guides, templates, and training curricula to realize its concepts [9, 26, 49, 73, 123]. Workshops applying the framework can not only provide concrete guidance for context-specific AI projects, they can also help iteratively refine the framework based on practitioner feedback.

A critical component of this refinement will be the development of education and training resources for both technologists and FCWs. Our framework is intended for use by technologists already familiar with facilitating participatory design workshops, but there remains room for tools and practices to sharpen and evaluate these competencies. Technologists' abilities here are critical, as they will be responsible for not only guiding the design process, but also equipping FCWs with the background needed to engage with the framework. As highlighted in the framework (Section 4), frontline care work contexts present a particular challenge because AI is not typically a core competency for either FCWs or their employers. This may contribute to extremes such as uncritical techno-optimism or self-doubt about engaging with data or AI systems, resulting in deference to others' expertise when interpreting or validating AI-enabled tools [117].

Future work should therefore examine what forms of AI understanding are minimally necessary for meaningful participation and for supporting FCWs in safeguarding themselves and their data during applied workshops, as well as how technologists might support this learning. As others have advocated [37], there are opportunities to develop stakeholder-first AI literacy workshops for FCWs and other stakeholders that emphasize reasoning about the implications of AI in specific care contexts, rather than focusing narrowly on technical skills. It is also important to consider how such education might be embedded within sociotechnical systems themselves [151], potentially expanding the scope of frontline care work to include AI literacy, and ensuring that FCWs are adequately trained and fairly compensated for these additional skills and responsibilities.

Lastly, applied workshops will enable concrete exploration of how our framework can translate to industrial settings for technology development. Technologists who may experience additional friction implementing participatory methods, such as engineers and product managers in large tech companies, may respond to the framework's provocations differently, revealing new tensions to explore. This builds on existing work that explores how participatory research can be translated into

practical, streamlined approaches in industry settings [34, 140]. Such efforts will enable concrete exploration of how unions, regulators, FCWs, and other key stakeholders can be more explicitly integrated into technology processes that still center FCWs' needs [9, 41, 73, 123].

*Expanding the Framework.* Applying the framework in real-world settings could also expand its breadth across geographies, domains, and stakeholder perspectives. Our work has focused on how to work with FCWs, as they are a critical yet under-resourced node in the care ecosystem, and under-studied in CSCW relative to care recipients like patients and consumers. However, care is socially complex, and we look forward to future work bringing a multi-stakeholder lens encompassing care recipients, employers, unions, and regulators. In addition, future work can use the framework in other CSCW research contexts in diverse geographic regions (e.g., in Global Majority settings [66]) or professional domains (e.g., with teachers [70], therapists [50], social workers [74], and others). Such applications could help surface both the “human universals” and “cultural differences” across frontline care work contexts [134], while also potentially revealing opportunities for worker solidarity [76]. Furthermore, the framework could be extended to workplace settings beyond frontline care work that still involve direct human-to-human connective labor, such as managerial roles [109]. Another potential expansion involves broadening the definition of care beyond interpersonal interactions to include non-human actors, technologies, and the surrounding environment—for example, in service-oriented domains like hospitality [120], the maintenance and repair of sociotechnical systems [30], or interactions with the more-than-human world [101].

*Iterating the Framework.* We also acknowledge that putting this conceptual framework into practice will inevitably reshape it. While we cannot anticipate precisely how, such changes could include revisions to the wording of the questions or shifts in emphasis across sections, while still preserving the framework's overall structure—as we observed when applying it to our case study examples. For instance, future applications might expand the meso-level factors that influence participation either by adding new considerations or by increasing the section size within the framework. Other adaptations could involve structural modifications, such as introducing additional stages in the participatory AI pipeline that may be more relevant to particular contexts. In this sense, the framework can be flexible to changes brought on by these different applications. At the same time, the core steps (i.e., considering scope and purpose, factors influencing participation, and participatory mechanisms) are likely to remain central. Nevertheless, as the framework is applied across a broader range of contexts and depths of engagement, even this underlying structure could evolve. Ultimately, we present this framework as a starting point and provocation, intended to spark ongoing conversation about what participatory AI could constitute in specific frontline care work contexts.

## 7 Conclusion

This paper presents a framework for examining some of the key considerations around developing an AI system in a frontline care work context. The framework builds upon a tradition of CSCW literature around participatory design, workplace technologies, and frontline care work—updating it based on the unique considerations presented by emerging AI systems. We provide a visual representation of the framework with real-world examples to help guide or orient technologists who are considering the benefits and harms of a proposed AI system, specific contextual influences on participation, and ultimately identify appropriate mechanisms for meaningfully engaging with FCWs and other stakeholders. We specifically discuss in further depth some of the challenges in these frontline care work contexts, including engaging with heterogeneous and low-wage workers, handling difficult to quantify elements of care, and ensuring accountability in a hierarchical setting. Furthermore, the framework raises key considerations for other scholars focused on participation,

work, and frontline care in AI, presenting opportunities for future breadth across settings and depth into robust participatory infrastructures.

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## A Text-Only (Non-Visual) Versions of the Worksheet

The following is textual representations of the worksheets prompts as a more linear list of questions. This representation lists out each of the Box 4.X from the worksheets in order and labels them similarly to the way they are referred to in the instructions (Section 4) and examples (Section 5). Each of the questions that require responses are indicated by a bullet point or sub-bullet point. The *example answers* for the case studies are indicated in italics and blue.

### A.1 Empty Worksheet (Building on Figure 1)

Participatory AI in Frontline Care Work

- What is a short description of your project?

#### 4.1 AI Intervention

##### 4.1.1 Goal of Applying AI

- What is your problem statement?
- Please select one of the following end goal(s): Understand work/workers, Facilitate or mediate work, Connect workers, Other (please specify)
- What are some of the existing approaches and intended improvements?

##### 4.1.2 Stakeholders

- Who are some of the key stakeholders in your context and their key goals or perspectives?
- What are some of their key interactions or conflicts?

##### 4.1.3 AI vs. Human Tradeoffs

- What are some of the tradeoffs between using an AI system versus maintaining existing approaches? Consider: Data representation of constructs; Comprehensive data vs. surveillance; Human vs. machine bias; Efficiency vs. individual burden

#### 4.2 Factors Influencing Participation

##### 4.2.1 Macro (societal)

- What are some of the macro (societal) factors influencing participation in your context? Consider: Devaluation of care work; History of research injustice; Techno-optimism

##### 4.2.2 Meso (organizational)

- What are some of the meso (organizational) factors influencing participation in your context? Consider: Employers, unions; Care organizations <> tech companies; Care organizations <> regulatory bodies

##### 4.2.3 Micro (individual)

- What are some of the micro (individual) factors influencing participation in your context? Consider: Financial and time constraints; AI literacy; Worker positionality

#### 4.3 Levels of Participation

- Who are the key stakeholders whose viewpoints you want to consider for participation?
- What is the overall approach you want to take in “Data Collection and Stewardship,” specifically for collecting, processing, and storing data?
  - What participation mechanisms make sense to implement for a tier 0 of participation to “inform” participants in a one way flow of information?
  - For the tier 1 of “consult” and inviting opinions of participants?
  - For the tier 2 of “involve” and have participants advise but not decide?
  - For the tier 3 of “collaborate” where participants are negotiating tradeoffs?
  - For the tier 4 of “empower” where participants have decision-making or managerial power?
- What is the overall approach you want to take in “Model Design and Development,” specifically for using data and making technical decisions?

- What participation mechanisms make sense to implement for a tier 0 of participation to “inform” participants in a one way flow of information?
- For the tier 1 of “consult” and inviting opinions of participants?
- For the tier 2 of “involve” and have participants advise but not decide?
- For the tier 3 of “collaborate” where participants are negotiating tradeoffs?
- For the tier 4 of “empower” where participants have decision-making or managerial power?
- What is the overall approach for “System Maintenance and Evaluation” for upholding performance guarantees and sunseting models?
  - What participation mechanisms make sense to implement for a tier 0 of participation to “inform” participants in a one way flow of information?
  - For the tier 1 of “consult” and inviting opinions of participants?
  - For the tier 2 of “involve” and have participants advise but not decide?
  - For the tier 3 of “collaborate” where participants are negotiating tradeoffs?
  - For the tier 4 of “empower” where participants have decision-making or managerial power?
- Based on the macro, meso, and micro elements identified, which of the participatory mechanisms are the most appropriate for your context?

## A.2 Case Study 1: Shift Matching in Home Care (Building on Figure 2)

### Participatory AI in Frontline Care Work

- What is a short description of your project? *AI to allocate shifts for home care workers*

#### 4.1 AI Intervention

##### 4.1.1 Goal of Applying AI

- What is your problem statement? *Shift matching takes a lot of time and energy from home care agency coordinators*
- Please select one of the following end goal(s): Understand work/workers, Facilitate or mediate work, Connect workers, Other (please specify) *Facilitate or mediate work*
- What are some of the existing approaches and intended improvements? *Currently matches are done by hand by admins based on their knowledge of the worker/patient schedules, preferences, needs, skills. AI could be more efficient / improve care*

##### 4.1.2 Stakeholders

- Who are some of the key stakeholders in your context and their key goals or perspectives? *Worker wants consistent/convenient shifts + give good care; Patient wants their care needs met; Employer wants all clients covered + profit*
- What are some of their key interactions or conflicts? *They all build long-term relationships and likely defer to client needs*

##### 4.1.3 AI vs. Human Tradeoffs

- What are some of the tradeoffs between using an AI system versus maintaining existing approaches? Consider: Data representation of constructs; Comprehensive data vs. surveillance; Human vs. machine bias; Efficiency vs. individual burden
  - *already isolated workers may lose the opportunity to check-in with a human*
  - *first-come-first-serve matches may bias towards “more online” workers, location biases, etc.*
  - *miss contextual knowledge / rapport (personality/style, empathy/compassion)*
  - *burden on workers to identify, escalate, and fix potential issues*

#### 4.2 Factors Influencing Participation

#### 4.2.1 Macro (societal)

- What are some of the macro (societal) factors influencing participation in your context? Consider: Devaluation of care work; History of research injustice; Techno-optimism
  - *bias towards quantifiable factors may further invisibilize emotional/relational contributions/expertise of workers*
  - *minoritized workers can again have their data / participation misused*

#### 4.2.2 Meso (organizational)

- What are some of the meso (organizational) factors influencing participation in your context? Consider: Employers, unions; Care organizations <> tech companies; Care organizations <> regulatory bodies
  - *existing union presence as a way for workers' voices to be collected / heard*
  - *home care agencies feel less confident disputing narratives from tech companies*
  - *employers may be incentivized to retaliate against workers*

#### 4.2.3 Micro (individual)

- What are some of the micro (individual) factors influencing participation in your context? Consider: Financial and time constraints; AI literacy; Worker positionality
  - *worker have little control over their fluctuating schedules*
  - *workers may feel the need to defer to patients during matching, further minimizing their voices*
  - *workers are not trained / compensated for AI work*

### 4.3 Levels of Participation

- Who are the key stakeholders whose viewpoints you want to consider for participation? *tech companies that are proposing this software, employers, home care workers*
- What is the overall approach you want to take in “Data Collection and Stewardship,” specifically for collecting, processing, and storing data? *using existing/future matches + preferences, satisfaction, needs, skills*
  - What participation mechanisms make sense to implement for a tier 0 of participation to “inform” participants in a one way flow of information? *employers are instructed (but don't give feedback) on how matches + data are collected*
  - For the tier 1 of “consult” and inviting opinions of participants? *interviews with workers/employers to understand data collected*
  - For the tier 2 of “involve” and have participants advise but not decide? *advisory board of workers and employers that can share perspectives on data collection + how to operationalize data*
  - For the tier 3 of “collaborate” where participants are negotiating tradeoffs? *workers select stewards (union, other workers, coordinators) to be involved in conversations about how to manage their data and its usage*
  - For the tier 4 of “empower” where participants have decision-making or managerial power? *power to decide what data is collected, veto usage, and what data is used for*
- What is the overall approach you want to take in “Model Design and Development,” specifically for using data and making technical decisions? *agency criteria + worker/ patient data -> ranked list of matches*
  - What participation mechanisms make sense to implement for a tier 0 of participation to “inform” participants in a one way flow of information? *model is developed/deployed on static datasets, no input beyond that*
  - For the tier 1 of “consult” and inviting opinions of participants? *employers and workers give feedback on the matches thru interviews + surveys*

- For the tier 2 of “involve” and have participants advise but not decide? *advisory board of workers and employers that can share perspectives on data collection + how to operationalize data*
- For the tier 3 of “collaborate” where participants are negotiating tradeoffs? *workers select stewards (union, other workers, coordinators) to be involved in conversations about how to manage their data and its usage*
- For the tier 4 of “empower” where participants have decision-making or managerial power? *cooperative ownership of the data used by the system and the model*
- What is the overall approach for “System Maintenance and Evaluation” for upholding performance guarantees and sunseting models? *periodic assessments of efficacy + model fine tuning*
  - What participation mechanisms make sense to implement for a tier 0 of participation to “inform” participants in a one way flow of information? *employers are informed about the results of periodic assessments*
  - For the tier 1 of “consult” and inviting opinions of participants? *survey for employers / workers to report issues*
  - For the tier 2 of “involve” and have participants advise but not decide? *workers and employers upvote/downvote on which issues they want addressed*
  - For the tier 3 of “collaborate” where participants are negotiating tradeoffs? *stewards periodically reassess system performance and update the design*
  - For the tier 4 of “empower” where participants have decision-making or managerial power? *total control over defining and addressing model performance*
- Based on the macro, meso, and micro elements identified, which of the participatory mechanisms are the most appropriate for your context?
  - **Tier 3 for Data and Model:** *workers select stewards (union, other workers, coordinators) to be involved in conversations about how to manage their data and its usage*
  - **Tier 1 for System:** *survey for employers / workers to report issues*

### A.3 Case Study 2: Burnout for Resident Physicians (Building on Figure 3)

#### Participatory AI in Frontline Care Work

- What is a short description of your project? *AI to identify burnout among medical residents based on multimodal data*

#### 4.1 AI Intervention

##### 4.1.1 Goal of Applying AI

- What is your problem statement? *medical residents are burned out and its hard to identify + intervene systematically*
- Please select one of the following end goal(s): Understand work/workers, Facilitate or mediate work, Connect workers, Other (please specify) *Understand work/workers*
- What are some of the existing approaches and intended improvements? *other contexts have used approaches like surveys + physiological measures; there is also research exploring how burnout can be identified from clinical notes*

##### 4.1.2 Stakeholders

- Who are some of the key stakeholders in your context and their key goals or perspectives? **Medical Resident** *wants to provide good care at minimal cost to self*; **Physician/Supervisor** *wants to train residents well and provide good care*; **Hospital Admin** *wants to make sure patients get care*

- What are some of their key interactions or conflicts? *the resident<->physician have a trainer/trainee relationship, the physician<->admin are focused on compliance; resident<->admin are based on support infrastructure*

#### 4.1.3 AI vs. Human Tradeoffs

- What are some of the tradeoffs between using an AI system versus maintaining existing approaches? Consider: Data representation of constructs; Comprehensive data vs. surveillance; Human vs. machine bias; Efficiency vs. individual burden
  - *residents may exhibit response bias (i.e., act differently because they are being observed) and skew the data*
  - *regulations (e.g., HIPAA) may restrict the type of data that could be used*
  - *difficult to balance figuring out who to target for accountability vs. retaliation*

### 4.2 Factors Influencing Participation

#### 4.2.1 Macro (societal)

- What are some of the macro (societal) factors influencing participation in your context? Consider: Devaluation of care work; History of research injustice; Techno-optimism
  - *qualitative / relational data / contributions may still be undervalued in comparison to more quantitative measures*

#### 4.2.2 Meso (organizational)

- What are some of the meso (organizational) factors influencing participation in your context? Consider: Employers, unions; Care organizations <-> tech companies; Care organizations <-> regulatory bodies
  - *insurance companies may feel the original purpose of clinical notes is diluted if the data is used in this new way*
  - *representational groups (e.g., ACGME) may not have insight into individual workplaces*

#### 4.2.3 Micro (individual)

- What are some of the micro (individual) factors influencing participation in your context? Consider: Financial and time constraints; AI literacy; Worker positionality
  - *difficult to engage workers already at high risk of burnout*
  - *getting baseline measurements for burnout requires continuous participation from residents*

### 4.3 Levels of Participation

- Who are the key stakeholders whose viewpoints you want to consider for participation? *researchers/technologists exploring this approach; supervisors/hospital admins; residents*
- What is the overall approach you want to take in “Data Collection and Stewardship,” specifically for collecting, processing, and storing data? *building on existing clinical note taking + feedback process*
  - What participation mechanisms make sense to implement for a tier 0 of participation to “inform” participants in a one way flow of information? *the system is designed, developed, and evaluated by the researchers / technologists and then residents / supervisors are instructed on how to use the systems*
  - For the tier 1 of “consult” and inviting opinions of participants? *residents/supervisors give feedback on the kinds of data that could be used*
  - For the tier 2 of “involve” and have participants advise but not decide? *residents develop paradigms around which data is used that are available to the system*
  - For the tier 3 of “collaborate” where participants are negotiating tradeoffs? *residents can set rules around which kinds of data are “donated” to the model*

- For the tier 4 of “empower” where participants have decision-making or managerial power? *residents have the ability to run the analysis process on their data when they want to*
- What is the overall approach you want to take in “Model Design and Development,” specifically for using data and making technical decisions? *training AI on diff sources of data to identify burnout*
  - What participation mechanisms make sense to implement for a tier 0 of participation to “inform” participants in a one way flow of information? *the system is designed, developed, and evaluated by the researchers / technologists and then residents / supervisors are instructed on how to use the systems*
  - For the tier 1 of “consult” and inviting opinions of participants? *residents are asked to rate the accuracy / appropriateness of potential outputs of the model*
  - For the tier 2 of “involve” and have participants advise but not decide? *residents/supervisors are brought together to discuss inputs/outputs to the model*
  - For the tier 3 of “collaborate” where participants are negotiating tradeoffs? *residents debate pros/cons of potential interpretations of the data from the model*
  - For the tier 4 of “empower” where participants have decision-making or managerial power? *residents can personalize how the model works for their own situations*
- What is the overall approach for “System Maintenance and Evaluation” for upholding performance guarantees and sunseting models? *following HIPAA (etc) guidelines on how frequently to review/update system*
  - What participation mechanisms make sense to implement for a tier 0 of participation to “inform” participants in a one way flow of information? *the system is designed, developed, and evaluated by the researchers / technologists and then residents / supervisors are instructed on how to use the systems*
  - For the tier 1 of “consult” and inviting opinions of participants? *residents/supervisors have the option to fill out a feedback from if issues arise*
  - For the tier 2 of “involve” and have participants advise but not decide? *focus groups are held periodically for feedback from residents/supervisors*
  - For the tier 3 of “collaborate” where participants are negotiating tradeoffs? *residents/supervisors help prioritize system issues*
  - For the tier 4 of “empower” where participants have decision-making or managerial power? *resident+supervisor council is responsible for ensuring the system continues to work*
- Based on the macro, meso, and micro elements identified, which of the participatory mechanisms are the most appropriate for your context?
  - **Tier 1 for Data:** *residents/supervisors give feedback on the kinds of data that could be used*
  - **Tier 2 for Model Design:** *residents/supervisors are brought together to discuss inputs/outputs to the model*
  - **Tier 3 for System:** *focus groups are held periodically for feedback from residents/supervisors*

B Initial Organization of Participation Considerations in Frontline Care Work Settings

| Challenges in FCW-centered participatory AI                                                                                                                                                                                                                                                                                                                                                             | Specific considerations for frontline care                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Approaches to addressing these challenges                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Participation Goal</b>                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                       |
| <p><i>What is the goal of applying AI or data-driven technologies in your intervention?</i></p> <ul style="list-style-type: none"> <li>What is your end goal, e.g., to understand work or workers, to facilitate or mediate work, or to connect workers?</li> <li>How would AI or data improve on what currently exists in the ecosystem to accomplish this goal—technical or non-technical?</li> </ul> | <ul style="list-style-type: none"> <li>Existing approaches in this field have centered knowledge workers (e.g., measuring productivity from device use). However, FCWs operate differently (e.g., moving between high- and low-connectivity, spending less time in front of digital devices, doing equal parts emotional and affective labor [138]). How might these realities make different end goals and AI applications more or less feasible?</li> </ul>                                                                                                                | <ul style="list-style-type: none"> <li>Discussing early on with community partners what the goal should be—there were a bunch of different goals and we needed to balance with the group’s existing efforts</li> </ul>                                                                                                                                |
| <p><i>Who are the stakeholders you will prioritize?</i></p> <ul style="list-style-type: none"> <li>What power dynamics exist between [workers and other] stakeholders and how can they be accounted for?</li> <li>How will you handle conflicts within and between stakeholders?</li> </ul>                                                                                                             | <ul style="list-style-type: none"> <li>FCWs are geographically distributed and work under strict hierarchies, making it difficult for them to connect and build representational power [111, 122].</li> <li>FCWs are often beholden not just to their employers’ demands, but also to their clients’ needs [43]. How might these [dependencies] be disentangled to understand workers’ [true] opinions?</li> </ul>                                                                                                                                                           | <ul style="list-style-type: none"> <li>Home care workers approached new data-driven technologies with almost unbridled optimism and had few privacy or security worries about sharing their own information if this helps with my patients. The only worries they had, at all, were for their clients’ information, rather than their own.</li> </ul> |
| <p><i>What are the limitations of applying AI and data-driven technology in this ecosystem?</i></p> <ul style="list-style-type: none"> <li>What is the history of introducing data-driven technologies into this ecosystem?</li> <li>How might these technologies impact accountability and responsibility for those in power?</li> </ul>                                                               | <ul style="list-style-type: none"> <li>FCWs’ voices have not historically been included in the design and development of the technologies they use, resulting in added operational burden [89].</li> <li>Historically, AI and data-driven technologies have inadequately captured emotional and relational aspects of care work [11]. This has allowed those in power to narrowly define which parts of workers’ contributions are counted and valued, and turned care work into data work [116, 119]. How does your technology reinforce or resist these trends?</li> </ul> | <ul style="list-style-type: none"> <li>It is important to examine the invisible work in a system in order to ensure contributions/concerns are not being overlooked [89]</li> </ul>                                                                                                                                                                   |

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## Participation Scope

### *How will the data be collected?*

- Will you use active or passive collection?
- What data type, and at what level of fidelity, is needed for the research question?
- FCWs may not be willing to do active data entry if it takes time away from the affective and emotional connections fundamental to care [68]. For some FCWs, however, contributing their oft-overlooked perspectives to research may be an exciting opportunity.
- The intimacy of frontline care means that any data are potentially revealing; however, FCWs often foreground the autonomy and privacy of their patients or clients alongside or even before their own [136].
- Offer FCWs multiple options for collecting data, at multiple modalities and fidelities, e.g., the burnout study paired structured burnout questionnaires, passively collected productivity data, and free-text qualitative responses contextualizing their productivity data.
- Give FCWs transparency and control over whether and how their data is collected, e.g., the options for manual or automatic uploading used in [the home care work tracking study], or other tools from data refusal [151]

### *How will the data be analyzed and stored?*

- Who is responsible for managing the data and how long should the data be stored (short-, medium- and long-term)?
- Who is responsible for analyzing the data, and how are participants involved (who should or should not have access)?
- If a model is created and deployed, how will it be audited and maintained?
- Frontline care workforces experience rapid turnover and resource constraints, making medium- or long-term data stewardship uniquely challenging.
- Regulatory standards for secure handling of frontline care data are not yet settled. The data might be health information, proof of productivity, reporting on occupational hazards, or research data. It is also unclear who the data belong to (e.g., the FCWs, their patients or clients, the FCWs' employers, or another party).
- FCWs often want their superiors to use the data to improve material conditions; this does, however, require superiors to have just enough information to spur action without compromising privacy.
- Use community governance structures like worker-led cooperatives to have FCWs decide how to manage the data. Note a FCWs-led decision-making process may decide to split responsibility between stakeholders, to limit burdens on FCWs and best use researchers' expertise [136].
- Consider tools for continuous participant interaction to validate the system and give dynamic consent, e.g., tools for individual workers to audit what their superior can see of their data, and to opt in or out [72].
- Carefully consider how much involvement in data stewardship activities imposes unwanted burdens on workers and stakeholders.

## Form of Participation

*What capacity do stakeholders have to participate?*

- What are external forces (i.e., institutional hierarchy, time/financial resources, AI literacy) that impact stakeholders' ability to participate?
- How do stakeholders' capacities change between the design/develop and deploy/maintain stages of a project?
- Existing frameworks often assume participation in AI takes significant computational and mathematical skill (e.g., coding, understanding probabilities). However, that puts an undue burden on FCWs to acquire these skills which may ultimately not be core to providing care and can actually distract from it.
- Often participatory methods assume that participants are agentic actors with meaningful control over their own schedules, in contrast technologists should consider that FCWs often tend to lack control over their own schedules as they must prioritize the needs of patients.
- Ground participation in concrete and actionable exercises that aim to address current problems
- Focus on stakeholder-first AI and technology literacy approaches that emphasize understanding AI implications over technological skills (e.g., coding)
- Design affordances the help facilitate gathering participants' perspectives and context on these systems and these data
- Consider new compensation models for participants involved in PD and longer-term relationships beyond standard research engagements

*What are the stakeholders' willingness to participate?*

- What would make the burdens of participation (e.g., affective and material [38]) worth it for a stakeholder?
- How much do stakeholders perceive their participation matters?
- How do you ensure that workers voices are heard and they benefit from these design considerations (i.e., increased respect, greater purpose, tools that better consider their needs, etc.)?
- What are the risks to participants abilities to earn a living (e.g., fired from their employer, membership revoked from union, loss of patients or clients, reduction in working hours, etc.)?
- Are their risks of being further devalued or de-prioritized (e.g., not assign patients or clients, removed from union activities, etc.)?
- What are the relationship dynamics between participants and technologists?
- Have you considered how the history of research injustice may impact participation?
- How do the social, cultural, and political contexts of these technologies influence participation?
- Take stock and reflect on participants' positionalities and personal experiences
- Support existing technological resources and ongoing grassroots community efforts
- Utilize and leverage workers collective voices (e.g., use aggregate data were applicable, work with unions and advocacy organizations, etc.)
- Challenge corporate design thinking and the primacy of quantitatively measured outcomes
- Outline and take measures to de-risk and mitigate potential negative outcomes (i.e., utilize university IRBs, consult community members and domain experts, etc.)
- Aim to understand the sociotechnical nature of these technologies